

Optimization Methods for Machine Learning and Engineering

Lecture 3 – Inequality Constrained Optimization

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Agenda

1. Some Notions of Topology
2. Inequality Constraints
3. The Interior-Point Method
4. Linear Programming
5. Soft Equality Constraints

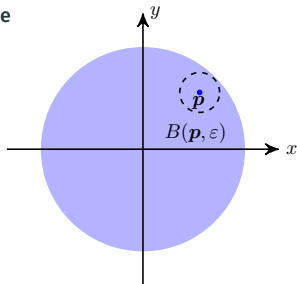
Some Notions of Topology

Open and Closed Sets (in Euclidean Spaces)

Open Sets

A set $P \subseteq \mathbb{R}^n$ is open iff
 $\forall \mathbf{p} \in P, \exists \varepsilon > 0, B(\mathbf{p}, \varepsilon) \subseteq P.$

Example

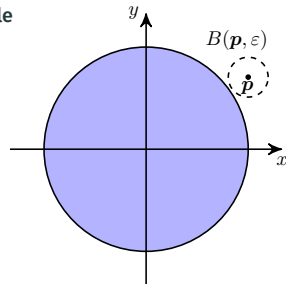


$$P = \{(x, y) \mid x^2 + y^2 < 1\}$$

Closed Sets

A set $P \subseteq \mathbb{R}^n$ is closed iff
 $\forall \mathbf{p} \in P^c, \exists \varepsilon > 0, B(\mathbf{p}, \varepsilon) \cap P = \emptyset.$

Example



$$P = \{(x, y) \mid x^2 + y^2 \leq 1\}$$

- The set $B(\mathbf{p}, \varepsilon) = \{\mathbf{x} \mid \|\mathbf{x} - \mathbf{p}\| < \varepsilon\}$ contains the points in a **ball with radius ε** around $\mathbf{p}$
- The set $P^c = \mathbb{R}^n \setminus P$ is called the **complement of P** and contains all points outside of P
- The **complement of an open set is closed** (and vice versa)

Properties of Open and Closed Sets

Convergent Sequences and Limit Points

A **closed set** P **contains its limit points**. That is, a convergent sequence in P converges to a point also in P .

For example the sequence $p_n = \frac{1}{n}$ is contained in $P = (0, 1]$. But it converges to $0 \notin P$.

$$p_1 = 1, \quad p_2 = \frac{1}{2}, \quad p_3 = \frac{1}{3}, \quad \dots$$

Unions and Intersection of Open Sets

- The **intersection of a finite number** of open sets is open
- The **union of any collection** of open sets is open

$$A_n = (-\frac{1}{n}, \frac{1}{n}), \quad \bigcap_{n=1}^{\infty} A_n = \{0\}$$

This intersection of infinitely many open sets
is closed!

Unions and Intersections of Closed Sets

- The **intersection of any collection** of closed sets is closed
- The **union of a finite number** of closed sets is closed

$$B_n = [\frac{1}{n}, 1 - \frac{1}{n}], \quad \bigcup_{n=2}^{\infty} B_n = (0, 1)$$

This union of infinitely many closed sets is open!

Closure and Interior

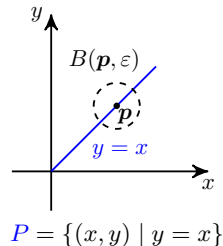
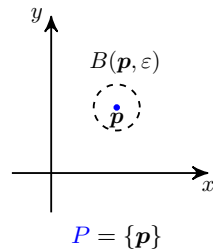
Open and Closed are not contradictory

- There are sets that are **neither** open nor closed.
For example $(0, 1]$.
- There are sets that are **both** open and closed.
For example the empty set $\emptyset = \{ \}$

Closure and Interior

- The **closure** of a set P is the smallest closed set that contains P .
- The **interior** of a set P is the biggest open set that is contained in P . This can also be the empty set if there is “no interior”.

Sets with Empty Interior



Inequality Constraints

Optimal Resource Allocation

Suppose you have **5 M€ to invest**. The money can be distributed to **n possible investments**. For example

- Startups
- Advertisement campaigns
- Energy-saving heating insulation

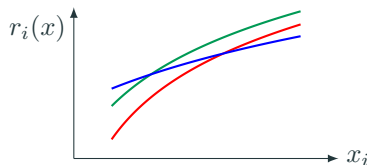
The **expected return** for each investment is described by a **concave reward function** r_i .

⇒ Putting more and more money into the same investment eventually has **diminishing returns** per invested Euro.

**How much to put into every investment
to maximize the overall reward?**

This is called the **Optimal Resource Allocation Problem**.

Concave Reward Functions



Optimization Problem **with Constraints**

$$\begin{aligned} & \max_{\mathbf{x} \in \mathbb{R}^n} && \sum_{i=1}^n r_i(x_i) \\ & \text{subject to} && \sum_{i=1}^n x_i \leq 5 \\ & && x_i \geq 0, \quad i = 1, \dots, n \end{aligned}$$

Constrained Optimization

So far, we have seen methods for *unconstrained convex optimization problems* of the form

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & x \in \mathbb{R}^n.\end{array}$$

We are now interested in *constrained convex optimization problems*

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & x \in X\end{array}$$

for a *convex domain* $X \subseteq \mathbb{R}^n$.

The main idea is to *transform constrained optimization problems into unconstrained problems*. These can then be solved with the algorithms from the first two lectures.

Equality and Inequality Constraints

Inequality Constraints

$$X = \{\mathbf{x} \in \mathbb{R}^n \mid g(\mathbf{x}) \leq 0\}$$

For the inequality constraint $g : \mathbb{R}^n \rightarrow \mathbb{R}$

Equality Constraints

$$Y = \{\mathbf{x} \in \mathbb{R}^n \mid h(\mathbf{x}) = 0\}$$

For the equality constraint $h : \mathbb{R}^n \rightarrow \mathbb{R}$

Multiple constraints can be combined, effectively forming the intersection of their *feasible solution sets*.

$$X \cap Y = \{\mathbf{x} \in \mathbb{R}^n \mid h(\mathbf{x}) = 0 \wedge g(\mathbf{x}) \leq 0\}$$

The *canonical form of a constrained optimization problem* is

$$\begin{array}{ll} \min_{\mathbf{x} \in \mathbb{R}^n} & f(\mathbf{x}) \\ \text{subject to} & g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \\ & h_j(\mathbf{x}) = 0, \quad j = 1, \dots, l \end{array}$$

This **Lecture 3** introduces *inequality constraints*. The next **Lecture 4** is on *equality constraints*.

The Interior-Point Method

Indicator Functions for Unconstrained Optimization

The **indicator function** of a set X is defined as $I_X(\mathbf{x}) = \begin{cases} 0, & \text{if } \mathbf{x} \in X \\ \infty, & \text{else.} \end{cases}$

With the indicator function, any **constrained optimization problem** $f : X \rightarrow \mathbb{R}$ (with $X \subseteq \mathbb{R}^n$) can be **stated as an unconstrained optimization problem** $\tilde{f}(\mathbf{x}) = f(\mathbf{x}) + I_X(\mathbf{x})$.

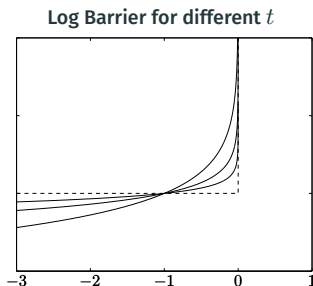
$$\begin{array}{ll} \text{minimize} & f(\mathbf{x}) \\ \text{such that} & \mathbf{x} \in X \end{array} \quad \Leftrightarrow \quad \begin{array}{ll} \text{minimize} & f(\mathbf{x}) + I_X(\mathbf{x}) \\ \text{such that} & \mathbf{x} \in \mathbb{R}^n \end{array}$$

- The unconstrained optimization problem has an extended image $\tilde{f} : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{\infty\}$
- If X is nonempty, then the optimizer of $\tilde{f}$ is also the optimizer of f
- But I_X is **non-differentiable on the closure of X** where optimal points can lie
 - So we cannot just use Gradient Descent or the Newton Method
 - Also, if the solution lies on the closure, we can have $\nabla \tilde{f}(\mathbf{x}^*) \neq \mathbf{0}$

(Logarithmic) Barrier Function

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$

such that $g(\mathbf{x}) \leq 0$



We approximate the indicator function $I_{\{\mathbf{x}|g(\mathbf{x}) \leq 0\}}$ with a **barrier function**. In this course we use the logarithmic barrier.

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) - \frac{1}{t} \log(-g(\mathbf{x}))$$

For $t \rightarrow \infty$ the barrier function $-\frac{1}{t} \log(-u)$ **approximates the indicator function** for values $u < 0$.

$$I_-(u) = \begin{cases} 0, & \text{if } u \leq 0 \\ \infty, & \text{else} \end{cases}$$

Since the barrier function is not defined for $u \geq 0$, the approximation is **valid only if we stay inside the barrier!**

Gradient and Hessian of the Barrier Function

Gradient and Hessian of the Logarithmic Barrier

$$\nabla \left[-\frac{1}{t} \log(g(\mathbf{x})) \right] = -\frac{1}{t} \frac{\nabla g(\mathbf{x})}{g(\mathbf{x})}$$

$$\nabla^2 \left[-\frac{1}{t} \log(g(\mathbf{x})) \right] = \frac{1}{t} \left(\frac{\nabla g(\mathbf{x}) \nabla g(\mathbf{x})^\top}{g(\mathbf{x})^2} - \frac{\nabla^2 g(\mathbf{x})}{g(\mathbf{x})} \right)$$

With this definition, we can simply **add the logarithmic barrier** to the **objective function, gradient and Hessian**. And then solve via Gradient Descent or the Newton Method.

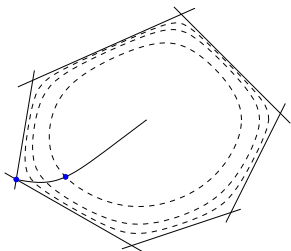
But we have to **start with an admissible point that fulfills all constraints**.

The **minimizer** of the approximated problem with the barrier **has a zero gradient** and fulfills all constraints.

“Looks like unconstrained optimization” because the algorithm iterations never leave the admissible region.

But how big should we set t for the barrier?

Sequential Unconstrained Optimization [Fiacco68]



Again, we have several inequality constraints $g_1, \dots, g_l$.

Increasingly **tighten the barrier** with $t_k > t_{k-1}$.

Solve numerically starting with the optimizer of the last iteration $\mathbf{x}^{k-1}$.

$$\mathbf{x}^k = \arg \min_{\mathbf{y} \in \mathbb{R}^n} \left[f(\mathbf{y}) - \frac{1}{t_k} \sum_{j=1}^l \log(-g_j(\mathbf{y})) \right] \quad (1)$$

Inner Iteration Steps of Gradient Descent / Newton Method to solve Equation (1) for a fixed t_k

Outer Iteration Increase $k \Rightarrow$ Tighten the barrier by selection of the next t_k

How fast to increase t_k

- Too fast: Large distance $\|\mathbf{x}^k - \mathbf{x}^{k-1}\| \Rightarrow$ No super-convergence, **many inner iterations** required
- Too slow: **Many outer iterations** required
- Theory of *self-concordant functions* increases t_k maximally fast for super-convergence [Nesterov94].
But the material is too advanced for this course.

Solving the Optimal Resource Allocation Problem

Suppose you have **5M€ to spend on advertisement** for your product. You want as many people as possible to have seen the product at least once.

TV Advertisement Costs 200k€ per run. 40 Mio people watch TV regularly. You have a 2% chance to be seen per TV watcher and ad run.

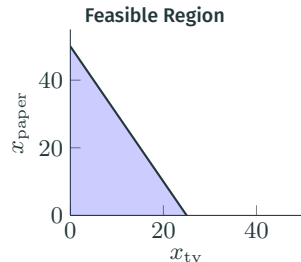
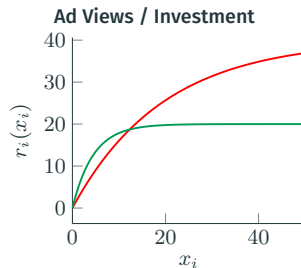
Newspaper Advertisement Costs 100k€ per run. 20 Mio people read newspapers regularly. You have a 20% chance per newspaper reader and ad run.

With x_{tv} , x_{paper} the number of ad runs, the number of people who saw the ad at least once is

$$r_{tv}(x_{tv}) = (1 - 0.98^{x_{tv}}) \cdot 40$$

$$r_{paper}(x_{paper}) = (1 - 0.8^{x_{paper}}) \cdot 20$$

The actual models of advertisement specialists are much more detailed than that. Even before hyper-targeted ads à la Facebook.



Solving the Optimal Resource Allocation Problem 2

Objective Function with Inequality Constraints

$$\max_{x \in \mathbb{R}^2} (1 - 0.98^{x_1}) \cdot 40 + (1 - 0.8^{x_2}) \cdot 20$$

$$\text{subject to } 0.2 \cdot x_1 + 0.1 \cdot x_2 \leq 5$$

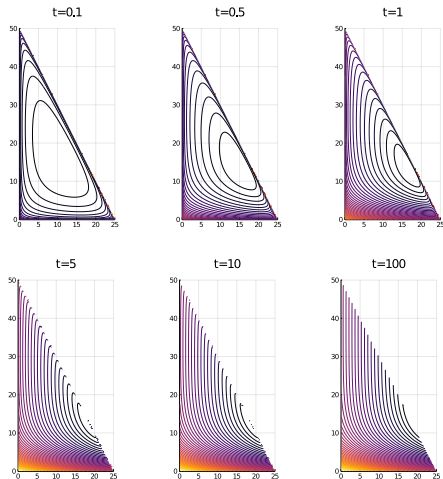
$$x_1 \geq 0, \quad x_2 \geq 0$$

Objective Function with Log Barriers

$$\begin{aligned} \min_{x \in \mathbb{R}^2} & - (1 - 0.98^{x_1}) \cdot 40 - (1 - 0.8^{x_2}) \cdot 20 \\ & - \frac{1}{t} \left[\log(5 - 0.2 \cdot x_1 - 0.1 \cdot x_2) + \right. \\ & \quad \left. \log(x_1) + \log(x_2) \right] \end{aligned}$$

The solution is $x^* = (18.768, 12.463)^\top$ with about 31.38 Mio people seeing the product at least once. This would also be a good **Mixed-Integer Problem** where the solution has to be a natural number. But we don't consider those in this course.

Contour-Lines of the Objective Function with Log Barriers



Finding an Admissible Initial Solution

The Interior-Point Method requires an **initial point** that is **strongly admissible** for all inequality constraints

$$g_i(\mathbf{x}^{\text{init}}) < 0, \quad i = 1, \dots, m$$

Finding Initial Interior Points

- Choose some $\mathbf{x}^0$ and then some s^0 so that $s^0 > \max(g_1(\mathbf{x}^0), g_2(\mathbf{x}^0), \dots)$
- $(\mathbf{x}^0, s^0)$ is an admissible point for the following (convex) optimization problem:

$$\begin{aligned} (\mathbf{x}^*, s^*) &= \arg \min_{\mathbf{x}, s} s \\ \text{subject to } & g_i(\mathbf{x}) - s \leq 0, \quad i = 1, \dots, m. \end{aligned} \tag{2}$$

Solve Equation (2) with the Interior-Point Method

- If $s < 0$, for some **intermediary solution** $(\mathbf{x}, s)$, then $\mathbf{x}$ is **strongly admissible (stop immediately)**
- If $s^* > 0$, then **no admissible solution exists**
- If $s^* = 0$, then $\mathbf{x}^*$ is an **admissible solution but cannot be used**
 - The Interior-Point Method is not suited for the original optimization problem
 - $\mathbf{x}^*$ is on the boundary, so the barrier function blows up immediately

Linear Programming

Motivation for Linear Programming

Stigler's Diet

- What is the cheapest combination of food that meets the requirements?
- Daily requirements for 9 different nutrients
- Nutrition content of 77 food types and their price
- Optimization in **77 dimensions** with **9 + 77 inequality constraints**
 - Fulfill the requirement in every category (9)
 - Only positive quantities for every food (77)

Linear Programming

- For large problems the notation for writing down all constraints separately becomes cumbersome
- How can we state the problem more compact?
- Standard form understood by software packages

TABLE 1. DAILY ALLOWANCES OF NUTRIENTS FOR A
MODERATELY ACTIVE MAN
(weighing 154 pounds)*

Nutrient	Allowance
Calories	3,000 calories
Protein	70 grams
Calcium	.8 grams
Iron	12 milligrams
Vitamin A	5,000 International Units
Thiamine (B ₁)	1.8 milligrams
Riboflavin (B ₂ or G)	2.7 milligrams
Niacin (Nicotinic Acid)	18 milligrams
Ascorbic Acid (C)	75 milligrams

* National Research Council, *Recommended Dietary Allowances*, Reprint and Circular Series No. 115, January, 1943.

Stigler's Solution

Food	Annual Quantities	Annual Cost
Wheat Flour	167.8 kg	\$13.33
Evaporated Milk	57 cans	\$3.84
Cabbage	50.3 kg	\$4.11
Spinach	10.4 kg	\$1.85
Dried Navy Beans	129.3 kg	\$16.80
Total Annual Cost		\$39.93

Stigler's Diet in Detail

Objective Function

We look for an solution $\mathbf{x} \in \mathbb{R}^{77}$ with a quantity for each of the 7 food types x_j for $j = 1, \dots, 77$.

The cost of each food type is known to be c_j . The column vector $\mathbf{c}$ contains the cost of all food types.

The **overall cost of a diet** $\mathbf{x}$ is then simply $\boxed{\mathbf{c}^\top \mathbf{x}}$.

Nutrient Requirement Constraints

The content of nutrients per food is stored in an individual column vector $\alpha_i \in \mathbb{R}^{77}$ per nutrient category i (calories, protein, etc.).

The requirement is to have at least β_i of each nutrient i . So the **nutrient constraints** are:

$$\boxed{\alpha_i^\top \mathbf{x} \geq \beta_i, \quad i = 1, \dots, 9}$$

Positivity Constraints

We cannot consume negative food. So we have to add **positivity requirements**:

$$\boxed{x_j \geq 0, \quad j = 1, \dots, 77}$$

What would happen without that constraint?

Affine Inequality Constraints

We are now in a more general setting (not only related to Stigler's Dient).

The individual affine inequality constraints can be **transformed** into the standard form $(\mathbf{a}_i^\top \mathbf{x} - b_i \leq 0)_{i=1,\dots,m}$ and **combined** to a single matrix $\mathbf{A}$ and vector $\mathbf{b}$.

$$\mathbf{A} = \begin{pmatrix} \text{---} \mathbf{a}_1^\top \text{---} \\ \text{---} \mathbf{a}_2^\top \text{---} \\ \vdots \\ \text{---} \mathbf{a}_m^\top \text{---} \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \Rightarrow X = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} - \mathbf{b} \leq \mathbf{0}\}$$

Given the above, the following **optimization problems are equivalent:**

$$\begin{array}{lll} \min_{\mathbf{x} \in \mathbb{R}^n} & f(\mathbf{x}) & \min_{\mathbf{x} \in \mathbb{R}^n} & f(\mathbf{x}) & \min_{\mathbf{x} \in X} & f(\mathbf{x}) \\ \text{subject to} & \mathbf{a}_1^\top \mathbf{x} - b_1 \leq 0 & \text{subject to} & \mathbf{A}\mathbf{x} - \mathbf{b} \leq \mathbf{0} & & \\ & \vdots & & & & \\ & \mathbf{a}_m^\top \mathbf{x} - b_m \leq 0 & & & & \end{array}$$

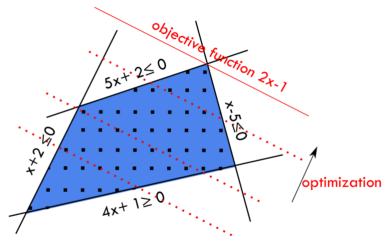
Linear Programming

$$\begin{array}{ll} \min_{x \in \mathbb{R}^n} & c^\top x \\ \text{subject to} & Ax - b \leq 0 \end{array}$$

Without the constraints, the linear target function $c^\top x$ would go to negative infinity.

Every constraint $a_i x - b_i \leq 0$ leaves **one side of a plane**, i.e. a **half-space**. The feasible solutions lie in the **intersection of the half-spaces**.

For a long time, it was believed that the boundary between efficiently solvable and effectively unsolvable problems was **linear vs. non-linear**. Today that perspective has shifted. The boundary is rather between **convex vs. non-convex problems**.



Why the Name Linear Programming?

The military refer to their various **plans or proposed schedules of training, logistical supply and deployment** of combat units as a program.

When I first analyzed the Air Force planning problem and saw that it could be formulated as a system of linear inequalities, I called my paper **Programming in a Linear Structure**.

Note that the term 'program' was used for linear programs **long before it was used as the set of instructions used by a computer**. In the early days, these instructions were called codes.

In the summer of 1948, Koopmans and I visited the Rand Corporation. One day we took a stroll along the Santa Monica beach. Koopmans said: **"Why not shorten 'Programming in a Linear Structure' to 'Linear Programming'?"** I replied: "That's it! From now on that will be its name."

George B. Dantzig. "Linear Programming". In: *Operations Research* 50.1 (2002), pp. 42–47

The Simplex Algorithm

Dantzig developed the **Simplex Algorithm** for solving LP during the Second World War. The main insight is that the **optimal solution will lie “on the edge”**. The intersection of two or more half-spaces. It was kept a **military secret** until 1947.

The Simplex Algorithm was widely used in economics [Koopmans51] and engineering. Kantorovich independently worked on LP and applied them to the war effort and economic planning in the UDSSR [Kantorovich39]. Koopmans and Kantorovich jointly won the **Nobel Price for Economics** in 1975.

The Simplex Algorithm performs good in practice. But it was later discovered, that edge cases can be constructed with **exponential $O(2^n)$ runtime** [Klee72]. Today's algorithms don't have this drawback.

The Simplex Algorithm is mostly of **historical interest**. We will not learn it in this course.

Simplex Algorithm “walking along the edges”

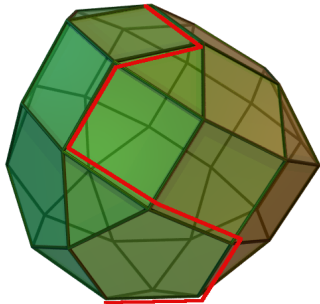


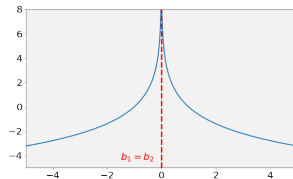
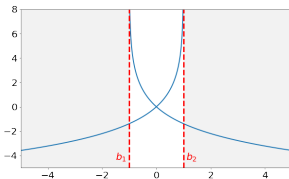
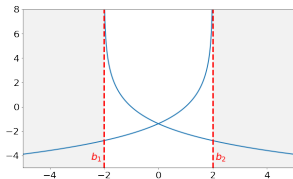
Image Source: Wikipedia

Soft Equality Constraints

Combine Inequality Constraints to get an Equality Constraint?

$$\begin{array}{ll} \min_{\mathbf{x} \in \mathbb{R}^2} & f(\mathbf{x}) \\ \text{subject to} & x_1 + x_2 = b \end{array} \quad \Leftrightarrow \quad \begin{array}{ll} \min_{\mathbf{x} \in \mathbb{R}^2} & f(\mathbf{x}) \\ \text{subject to} & x_1 + x_2 \geq b \\ & x_1 + x_2 \leq b \end{array}$$

- Both are “in principle” equivalent.
- But the set of admissible solutions in $\mathbb{R}^2$ has **no interior!**
- The solutions lie on a 1D plane embedded in $\mathbb{R}^2$.
- So we cannot use the Interior Point Method to solve the problem as stated on the right-hand side.



Soft Constraints

Suppose an optimization with an equality constraint

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & h(\mathbf{x}) = 0 \end{aligned}$$

A trivial way to transform it into an unconstrained optimization problem is to replace the hard equality constraint with a **soft constraint** in the form of a **penalty term added to the optimization function**.

For example as quadratic penalties weighted by a factor α :

$$\min_{\mathbf{x} \in \mathbb{R}^n} [f(\mathbf{x}) + \alpha h(\mathbf{x})^2]$$

Attention! **Soft constraints introduce a bias to the position of the optimal solution.** They are a **heuristic approximation** that can work well in practice. Further analysis is required to understand the introduced bias.

Notice the similarity between soft constraints and the regularization terms to prevent overfitting from Lecture 2.

Least-Squares Fitting of an Ellipse

x	y
1	7
2	6
5	8
7	7
9	5
6	7
3	2
8	4

- Due to gravity, *planets are (circa) on an elliptic orbit* around the sun
- All *points p exactly on an ellipse* obey the equation

$$\mathbf{p}^\top \mathbf{A} \mathbf{p} + \mathbf{b}^\top \mathbf{p} + c = 0 \quad (3)$$

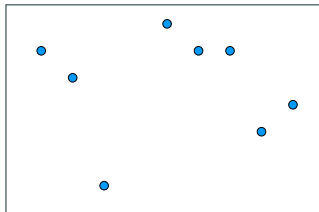
with $\mathbf{A}$ a symmetric matrix

- Suppose the simple case of an ellipse in the 2D-plane

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

- There are 6 unknown parameters $\boldsymbol{\theta} = (a_{11}, a_{12}, a_{22}, b_1, b_2, c)$

How can we fit these parameters to match observations?



Least-Squares Fitting of an Ellipse 2

Let $p \in D$ the observed points. The ellipse parameters are fitted with a least-squares objective function:

$$\min_{\theta=(a_{11}, a_{12}, a_{22}, b_1, b_2, c)} \sum_{p \in D} \underbrace{\left[\begin{pmatrix} p_1^2 & 2p_1p_2 & p_2^2 & p_1 & p_2 & 1 \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{12} \\ a_{22} \\ b_1 \\ b_2 \\ c \end{pmatrix} \right]}_{\text{Left-hand-side of Equation (3)}}^2$$

Note that $\theta = 0$ is a *trivial minimizer*. And very much uninformative.

For a (hypothetical) perfect fitting solution $\theta^+ \neq 0$, the left-hand-side of Equation (3) is exactly zero.

Every *scaled solution* $\alpha\theta^+$ would also result in a “perfect fit”. So the θ can be scaled arbitrarily.

We “arbitrarily” add $\theta_1 = a_{11} = 1$ as a *soft-constraint*. This *avoids the trivial solution* $\theta = 0$.

The other elements of θ will scale accordingly to minimize the overall error.

Least-Squares Fitting of an Ellipse 3

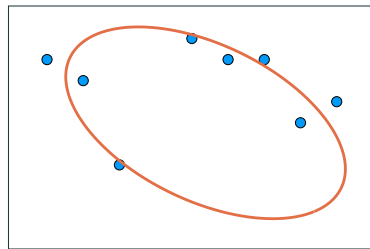
$$\min_{\theta=(a_{11},a_{12},a_{22},b_1,b_2,c)} \sum_{p \in D} \left[\begin{pmatrix} p_1^2 & 2p_1p_2 & p_2^2 & p_1 & p_2 & 1 \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{12} \\ a_{22} \\ b_1 \\ b_2 \\ c \end{pmatrix} \right]^2 + \underbrace{\beta (a_{11} - 1)^2}_{\text{Penalty Term for the Soft Constraint}}$$

β is selected very large to ensure that $a_{11} \approx 1$. But even for large β the **solution does not obey the soft-constraint perfectly**.

For $\beta = 10^4$ we get $\theta^* = \begin{pmatrix} 0.99179 \\ 0.40439 \\ 0.71326 \\ -13.90265 \\ -10.05387 \\ 46.12138 \end{pmatrix}$.

In the next lecture we learn how **hard equality constraints** can be formulated that are obeyed perfectly by the solution (if a solution exists).

Fitted Ellipse



Summary of what you learned today

- The definition of **open and closed sets**, as well as the **interior** of a set.
- The difference between **constrained and unconstrained** optimization problems.
 - Inequality constraints
 - Equality constraints
- The **Interior Point Method** used to solve problems with inequality constraints
 - How a **logarithmic barrier** is constructed for inequality constraints
 - How an **initial interior point** can be found that obeys all inequality constraints
- **Linear Programming**, a common category of optimization problems
- The use of **soft constraints** to approximate equality constraints

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